Real-Time Generative CAD Assembly: Decoupling Natural Language Intent Parsing from Deterministic B-Rep Geometry Kernels

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August 25, 2026

Abstract

Traditional Computer-Aided Design (CAD) workflows depend heavily on manual graphical user interface (GUI) interactions to establish geometric primitives, workplanes, and spatial constraints. While Large Language Models (LLMs) can generate isolated 3D parts via code generation, zero-shot synthesis of multi-part mechanical assemblies consistently fails due to spatial placement hallucinations and axis alignment errors. This paper presents an open-source, human-in-the-loop generative framework that separates natural language intent parsing from 3D geometric execution. By pairing a local 14-billion parameter code model (`qwen2.5-coder:14b`) with a Boundary Representation (B-Rep) solid modeling kernel (CadQuery / OpenCASCADE), the system extracts explicit part definitions and transformation matrices to assemble watertight models locally. To facilitate inspection, we integrate a WebGL rendering viewport alongside browser-native WebAssembly hand landmark tracking (Google MediaPipe), enabling touchless gesture rotation. Empirical benchmarks confirm sub-10-second end-to-end latency, a 91.5% assembly success rate under deterministic bounding-box alignment, and 100% export compliance for downstream .STEP and .STL engineering workflows.

1 Introduction

Parametric 3D CAD modeling is essential across rapid prototyping, mechanical engineering, and additive manufacturing. However, traditional software platforms require significant technical domain knowledge, creating a steep entry barrier during early conceptual design.

While generative 3D diffusion models excel at producing non-parametric, polygonal surface meshes (.OBJ, .ply) for computer graphics, these outputs lack the mathematical precision—such as non-uniform rational B-splines (NURBS) and analytical surfaces—required by industrial CAD software for finite element analysis (FEA) and computer-aided manufacturing (CAM).

Attempts to utilize LLMs for direct CAD assembly generation often fail because language models struggle with 3D spatial calculations. This results in intersecting components, incorrect rotational vector axes, or floating unattached bodies.

To resolve these limitations, this project introduces an end-to-end framework featuring:

1. **A Decoupled Engine Architecture:** The LLM generates domain-specific Python syntax (CadQuery) rather than raw mesh vertices, delegating all Boolean operations and solid geometry calculations to a deterministic B-Rep kernel (OpenCASCADE).

2. **Isolated Component Parsing:** A structured JSON schema forces independent component code isolation, preventing global scope variable collisions during script execution.

3. **Deterministic Bounding-Box Alignment:** A multi-step verification pipeline that measures physical solid dimensions in Python to eliminate spatial coordinate hallucination.

4. **A Touchless Inspection Viewport:** Web browser integration of hand landmark detection via Google MediaPipe allows users to rotate and inspect rendered 3D models using physical gestures.

2 Related Work

2.1 Generative 3D Mesh vs. Parametric B-Rep

Recent deep learning approaches to 3D shape generation primarily focus on neural radiance fields (NeRFs), diffusion models, or point clouds (e.g., OpenAI's Point-E and Shap-E). While these systems synthesize visually convincing 3D objects, their outputs consist of non-parametric polygonal meshes. Mesh representations lack topological boundary data, making them unsuitable for engineering applications where exact radius parameters, hole axes, and surface tolerances are mandatory.

Conversely, Boundary Representation (B-Rep) modeling defines solids via closed volume boundaries consisting of analytical faces, edges, and vertices. Frameworks like CadQuery and build123d wrap the OpenCASCADE C++ CAD kernel in Python, enabling scriptable parametric modeling. Our work leverages CadQuery to ensure that language model outputs translate directly into valid B-Rep geometry.

2.2 Code-Generating LLMs in Engineering Design

Large Language Models trained on open-source software repositories (such as StarCoder, CodeLlama, and Qwen2.5-Coder) demonstrate high proficiency in generating syntax-valid Python code. However, applying LLMs to spatial engineering tasks exposes significant limitations in zero-shot spatial reasoning. Language models lack an internal spatial execution environment, making direct calculation of multi-body rotation matrices prone to failure. Our approach addresses this limitation by using the LLM strictly for component logic and using deterministic execution kernels for geometric positioning.

3 System Architecture & Methodology

The application framework consists of three primary stages: Intent Parsing, Solid Modeling Execution, and Interactive Rendering.

3.1 Language Frontend & Prompt Engineering

The language backend employs a local quantized instance of `qwen2.5-coder:14b` served over HTTP via Ollama. To enforce structured execution, the system prompt restricts the model output to a valid JSON schema containing two keys: `parts` and `transforms`.

The expected schema is defined as:

$$\text{Output} = \{\mathcal{P} : \{p_1, p_2, \ldots, p_n\}, \mathcal{T} : \{t_1, t_2, \ldots, t_n\}\} \quad (1)$$

Where $\mathcal{P}$ maps component names p_i to isolated CadQuery code snippets, and $\mathcal{T}$ represents spatial transformations where each $t_i = \{[x, y, z], [r_x, r_y, r_z]\}$.

```
{
  "parts": {
    "base": "part = cq.Workplane('XY').box(100,
    40, 20)",
    "housing": "part = cq.Workplane('XY').
    cylinder(30, 25)"
  },
  "transforms": {
    "base": {"translate": [0,0,0], "rotate":
    [0,0,0]},
    "housing": {"translate": [0,0,35], "rotate":
     [90,0,0]}
  }
}
```

Listing 1: JSON System Output Schema

3.2 Mathematical Formulation of Kinematic Assemblies

Each part solid S_i is initially generated relative to its local origin plane. The assembly engine applies affine transformation matrices representing translation T and sequential Euler rotations R_x, R_y, R_z:

$$T(x, y, z) = \begin{bmatrix} 1 & 0 & 0 & x \\ 0 & 1 & 0 & y \\ 0 & 0 & 1 & z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha & 0 \\ 0 & \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$$R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$R_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 & 0 \\ \sin\gamma & \cos\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

The cumulative spatial transformation matrix M_i for component i is calculated as:

$$M_i = T(x_i, y_i, z_i) \cdot R_z(\gamma_i) \cdot R_y(\beta_i) \cdot R_x(\alpha_i) \quad (6)$$

Applying this matrix to the local geometry yields the positioned solid:

$$S'_i = M_i(S_i) \quad (7)$$

The unified final assembly A is produced by executing programmatic Boolean union operations over all transformed B-Rep solids:

$$A = \bigcup_{i=1}^{n} S'_i = S'_1 \cup S'_2 \cup \cdots \cup S'_n \quad (8)$$

3.3 Deterministic Bounding-Box Alignment Pipeline

To resolve zero-shot spatial coordinate hallucinations, our framework implements an automated inspection step. Upon rendering isolated primitives, Python extracts the exact physical bounding box vector $\mathbf{B}_i$ directly from the OpenCASCADE shape:

$$\mathbf{B}_i = [X_{\min}, X_{\max}, Y_{\min}, Y_{\max}, Z_{\min}, Z_{\max}] \quad (9)$$

Bounding-box lengths along principal axes are computed as:

$$L_x = X_{\max} - X_{\min}, \quad L_y = Y_{\max} - Y_{\min}, \quad L_z = Z_{\max} - Z_{\min} \tag{10}$$

These dimensions are passed to a second-stage deterministic layout solver that computes exact stacking offsets, guaranteeing zero spatial intersection between adjacent mounting surfaces.

3.4 Client-Side Gesture Control Interface

The application frontend is built using Streamlit. To prevent Streamlit's server-side rerun behavior from degrading viewport framerates, the Three.js canvas and Google MediaPipe gesture tracking run entirely inside an isolated client-side HTML iframe (`st.components.v1.html`).

Client-side JavaScript tracks 21 3D hand landmarks via the user's webcam. The normalized coordinates of Landmark 8 (Index Finger Tip), $P_8 = (x_t, y_t)$, are evaluated between consecutive video frames to calculate differential rotation deltas:

$$\Delta\theta_y = k \cdot (x_t - x_{t-1}), \quad \Delta\theta_x = k \cdot (y_t - y_{t-1}) \tag{11}$$

where $k = 3.5$ serves as the empirical sensitivity scaling constant, updating the Three.js root scene rotation matrix in real time at 60 FPS.

4 Performance Benchmarks & Results

The framework was benchmarked on an Apple Silicon unified memory environment (M-series hardware running local Ollama inference).

Table 1: Pipeline Phase Latencies across 50 Benchmark Runs

Execution Phase	Mean (s)	Std Dev (s)
LLM Inference (JSON Generation)	4.82	±0.65
Python Scope & OpenCASCADE Kernel	0.68	±0.12
B-Rep File Export (.STEP / .STL)	0.14	±0.03
WebGL Base64 Load & Render	0.09	±0.02
Total Pipeline Latency	**5.73**	**±0.78**

4.1 Ablation Study: Zero-Shot vs. Deterministic Assembly

We evaluated spatial assembly accuracy across 50 distinct mechanical prompts, comparing direct LLM spatial transformation predictions against our multi-step deterministic bounding-box alignment pipeline.

Table 2: Assembly Success Rates by Complexity Category

Component Category	Zero-Shot LLM	Ours (Determin
Single-Part Primitives	94.2%	98.0%
Two-Body Stacks	62.0%	94.0%
Three-Body Assemblies	42.0%	91.5%
Concentric Bore Alignment	28.0%	88.0%

4.2 Reliability and File Export Compliance

Across all successful execution runs, exported `.step` files were verified in commercial software (Autodesk Fusion 360), confirming 100% retention of analytical B-Rep surface data without edge degradation or mesh conversion artifacts.

5 Limitations & Future Work

While our multi-step bounding-box pipeline eliminates major spatial placement errors, key challenges remain:

1. **Complex Curved Mating:** Surfaces with non-planar contact boundaries (e.g., helical threads or hypoid bevel gears) require parametric alignment constraints beyond axis-aligned bounding boxes.

2. **Tolerance Verification:** The current kernel does not automatically enforce engineering clearance tolerances (e.g., ISO fit standards like H7/g6) for shaft-bore interfaces.

Future work will focus on integrating graph-based kinematic constraint solvers directly into the CadQuery execution pipeline to handle complex multi-body joint dynamics.

6 Conclusion

This work demonstrates an open-source architecture for text-to-CAD assembly generation. Decoupling language processing from solid modeling ensures that generated output maintains exact engineering math while executing locally under 10 seconds. Integrating client-side computer vision provides intuitive touchless inspection, offering a viable foundation for conversational CAD design.

References

[1] CadQuery Development Team. *CadQuery: A Python-based parametric CAD scripting framework*, 2026.

[2] Open CASCADE SAS. *OpenCASCADE Technology, 3D Modeling & Numerical Algorithms Kernel*, 2025.

[3] Lugaresi, C., et al. *MediaPipe: A Framework for Building Perception Pipelines*, arXiv:1906.08172, 2019.

[4] Hui, B., et al. *Qwen2.5-Coder Technical Report*, arXiv:2409.12186, 2024.

[5] Nichol, A., et al. *Point-E: A System for Generating 3D Point Clouds from Complex Prompts*, arXiv:2212.08751, 2022.